

Configuration de l'autorité de certification

Sur SE4AD :

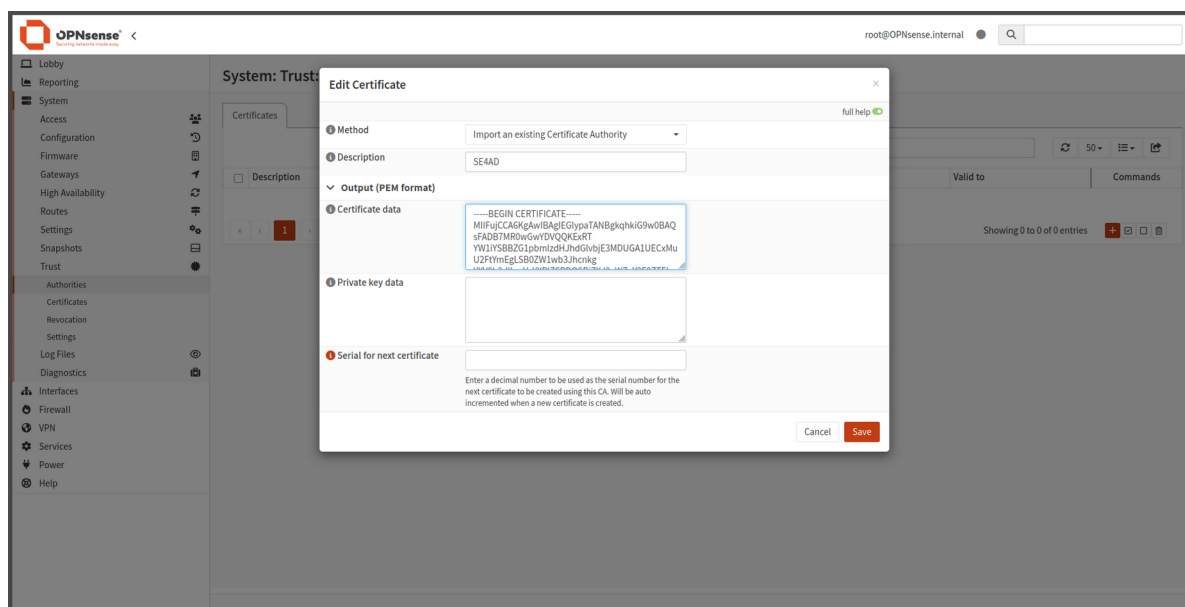
cat /var/lib/samba/private/tls/ca.pem

Sur OPNSense

Cliquer sur System → Trust → Authorities puis sur le + pour Ajouter une autorité connue.

Sélectionner la Méthode : « Import an existing Certificate Authority », donner une description.

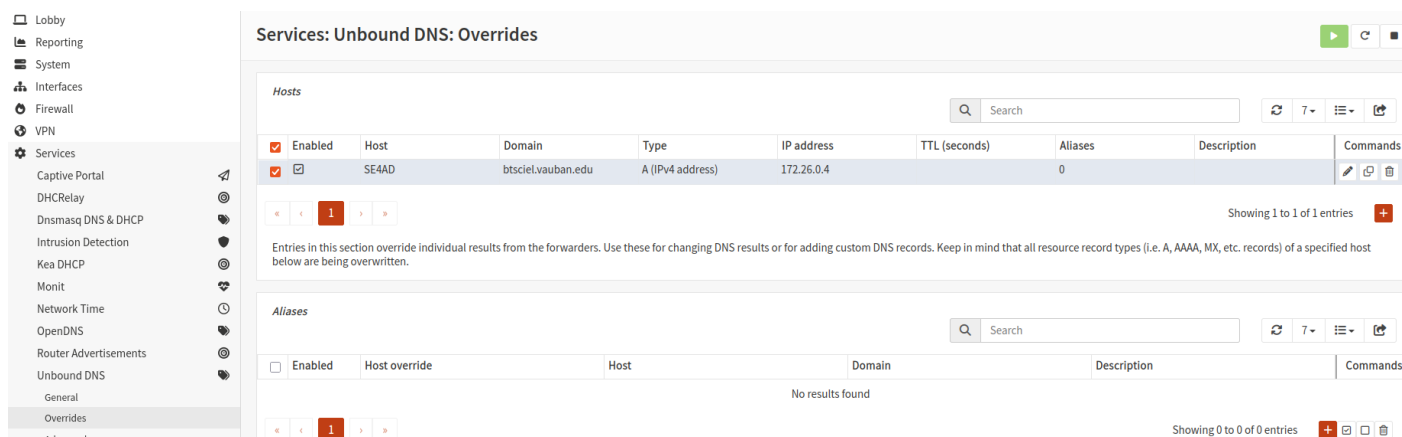
Copier ensuite le certificat (De -----BEGIN CERTIFICATE jusqu'à END CERTIFICATE-----) dans la Partie « Certificate Data » de l'OPNSense



Configuration de l'Unbound DNS

Il faut à présent indiquer à OPNSense la correspondance entre le nom DNS du serveur LDAP et son IP, cela se fait dans : **Services** → **Unbound DNS** → **Overrides** puis + **sur Hosts**

Ici : Host : SE4AD Domain : btsciel.vauban.edu IP address : 172.26.0.4



The screenshot shows the OPNSense web interface for configuring Unbound DNS overrides. The left sidebar lists various services, with 'Unbound DNS' selected. The main panel is titled 'Services: Unbound DNS: Overrides' and contains two sections: 'Hosts' and 'Aliases'.

The 'Hosts' section displays a table with the following columns: Enabled, Host, Domain, Type, IP address, TTL (seconds), Aliases, Description, and Commands. A single entry is shown:

Enabled	Host	Domain	Type	IP address	TTL (seconds)	Aliases	Description	Commands
☒	SE4AD	btsciel.vauban.edu	A (IPv4 address)	172.26.0.4		0		

Below the table, it indicates 'Showing 1 to 1 of 1 entries'.

The 'Aliases' section is currently empty, showing 'No results found'.

N'oubliez pas de sauvegarder ET Appliquer la configuration

Vous pouvez vérifier la bonne configuration en essayant de ping le nom DNS du serveur LDAP dans le shell opnsense.

```
0) Logout
1) Assign interfaces
2) Set interface IP address
3) Reset the root password
4) Reset to factory defaults
5) Power off system
6) Reboot system
7) Ping host
8) Shell
9) pfTop
10) Firewall log
11) Reload all services
12) Update from console
13) Restore a backup

Enter an option: 8

root@OPNsense:~ # ping SE4AD.btsciel.vauban.edu
PING SE4AD.btsciel.vauban.edu (172.26.0.4): 56 data bytes
64 bytes from 172.26.0.4: icmp_seq=0 ttl=64 time=1.141 ms
64 bytes from 172.26.0.4: icmp_seq=1 ttl=64 time=1.134 ms
^C
--- SE4AD.btsciel.vauban.edu ping statistics ---
2 packets transmitted, 2 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 1.134/1.138/1.141/0.004 ms
```

Configuration du serveur LDAP

Dans **System** → **Access** → **Servers** cliquer sur + **Add**

Remplir les champs comme dans la capture d'écran (en adaptant si besoin les noms)

Pour trouver/vérifier le User DN :

Sur le se4AD :

Taper :

```
ldbsearch -H /var/lib/samba/private/sam.ldb "(cn=admin)" dn
```

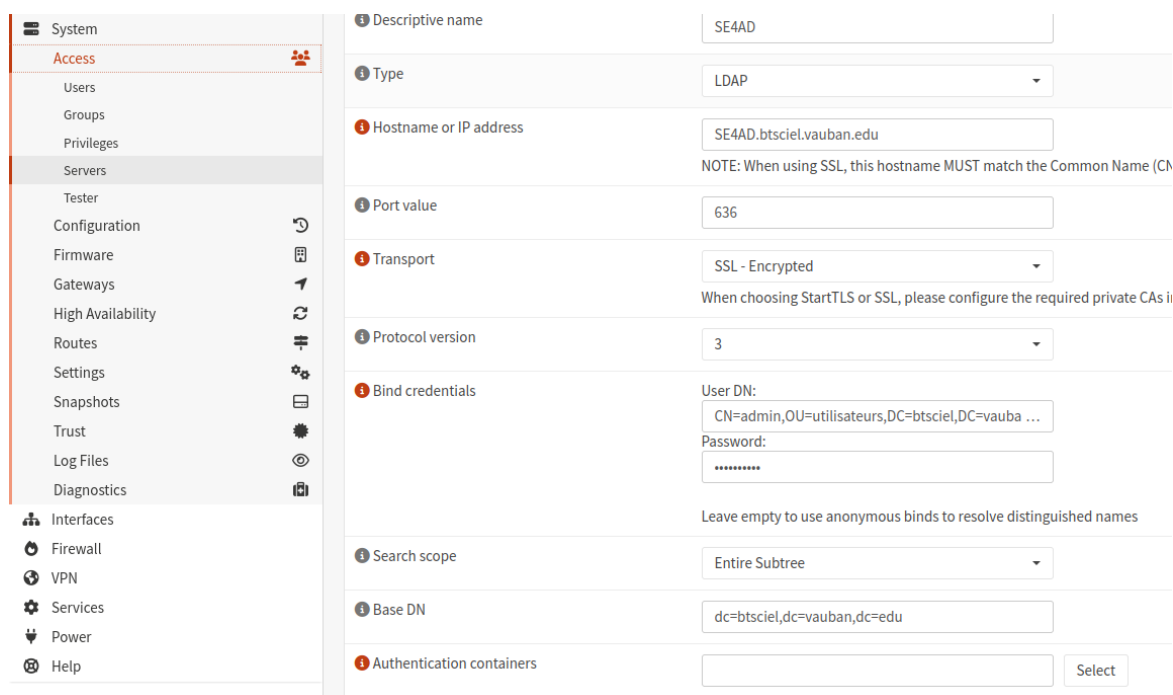
Cela doit vous renvoyer par exemple :

record 1

dn: CN=admin,OU=utilisateurs,DC=btsciel,DC=vauban,DC=edu

Copier/coller alors à partir de CN= dans le champ User DN de Bind credentials

Le mot de passe est le mot de passe admin de l'utilisateur Active Directory



The screenshot shows the OPNSense configuration interface for adding a new LDAP server. The left sidebar has a menu with 'System' expanded, showing 'Access' and 'Servers' (selected). The main panel is titled 'Descriptive name' and contains the following fields:

- Descriptive name:** SE4AD
- Type:** LDAP
- Hostname or IP address:** SE4AD.btsciel.vauban.edu
NOTE: When using SSL, this hostname MUST match the Common Name (CN)
- Port value:** 636
- Transport:** SSL - Encrypted
When choosing StartTLS or SSL, please configure the required private CAs in the Settings tab
- Protocol version:** 3
- Bind credentials:**
 - User DN: CN=admin,OU=utilisateurs,DC=btsciel,DC=vauba ...
 - Password: [masked]
- Search scope:** Entire Subtree
- Base DN:** dc=btsciel,dc=vauban,dc=edu
- Authentication containers:** [empty] Select

Cocher également les cases suivantes et remplir le « User naming attribute » : cn

Extended Query		Example: &(objectClass=inetOrgPerson)(mail="*@example.com)
User naming attribute	cn	Typically "cn" (OpenLDAP, Novell eDirectory), "sAMAccountName" (Microsoft AD)
Read properties	☒	Normally the authentication only tries to bind to the remote server, when this option is enabled also the objects properties are fetched, can be practical for debugging purposes.
Synchronize groups	☒	Synchronize groups specified by memberOf or class attribute after login, this option requires to enable read properties. Groups will be extracted from the first CN= section and will only be considered when already existing in OPNSense. Group memberships will be persisted in OPNSense. Use the server test tool to check if memberOf is returned by your LDAP server before enabling.
Default groups	Nothing selected	Group(s) to add by default when creating users
Constraint groups	☐	Constraint allowed groups to those selected in the container section. This may offer additional security in cases where users are able to inject memberOf attributes in different trees.
Limit groups (Scope)	Nothing selected	Limit the groups which may be used by this authenticator, keep empty to consider all local groups in OPNSense. When groups are selected, you can assign unassigned groups to the user manually
Automatic user creation	☒	To be used in combination with synchronize or default groups, allow the authenticator to create new local users after successful login with group memberships returned for the user.
Match case insensitive	☒	Allow mixed case input when gathering local user settings.
Save		

Cliquer ensuite sur « SELECT » en face de Authentication Container :

Authentication containers		Select
Semicolon-separated list of distinguished names containing DC= components. Example: OU=Freelancers,O=Company,DC=example,DC=com;CN=Users,OU=Staff,O=Company,DC=example,DC=com		

Si votre configuration est correcte une fenêtre doit s'ouvrir avec la liste des container LDAP :

Sélectionner alors

OU=utilisateurs,DC=btsciel,DC=vauban,DC=edu

Please select which containers to Authenticate against:

☐ OU=matieres,DC=btsciel,DC=vauban,DC=edu
☐ OU=cours,OU=Groups,DC=btsciel,DC=vauban,DC=edu
☐ OU=Administratifs,OU=utilisateurs,DC=btsciel,DC=vauban,DC=edu
☒ OU=utilisateurs,DC=btsciel,DC=vauban,DC=edu
☐ OU=Autres,OU=Groups,DC=btsciel,DC=vauban,DC=edu
☐ CN=Users,CN=Builtin,DC=btsciel,DC=vauban,DC=edu
☐ OU=equipements,OU=Groups,DC=btsciel,DC=vauban,DC=edu
☐ OU=Eleves,OU=utilisateurs,DC=btsciel,DC=vauban,DC=edu
☐ OU=classes,OU=Groups,DC=btsciel,DC=vauban,DC=edu
☐ OU=projets,OU=Groups,DC=btsciel,DC=vauban,DC=edu
☐ OU=Rights,DC=btsciel,DC=vauban,DC=edu
☐ OU=Profs,OU=utilisateurs,DC=btsciel,DC=vauban,DC=edu
☐ OU=Trash,DC=btsciel,DC=vauban,DC=edu
☐ CN=Users,DC=btsciel,DC=vauban,DC=edu
☐ OU=Equipements,DC=btsciel,DC=vauban,DC=edu
☐ OU=Domain Controllers,DC=btsciel,DC=vauban,DC=edu
☐ OU=Groups,DC=btsciel,DC=vauban,DC=edu
☐ OU=computers,DC=btsciel,DC=vauban,DC=edu
☐ OU=Materiels,DC=btsciel,DC=vauban,DC=edu
☐ OU=delegations,DC=btsciel,DC=vauban,DC=edu
☐ OU=etablisements,DC=btsciel,DC=vauban,DC=edu
☐ OU=Parcs,DC=btsciel,DC=vauban,DC=edu

Cliquer sur Save puis encore Save

En cas de message d'erreur, consultez les logs dans : « **System → Log files → General** »

Date	Severity	Process	Line
2026-03-05T13:39:19	Error	opnsense	LDAP bind error [80090308: LdapErr: DSID-0C0903A9, comment: AcceptSecurityContext error, data 52e, v1db1; Invalid credentials]
2026-03-05T13:39:15	Notice	configctl	event @ 1772717954.31 exec: system event config_changed response: OK
2026-03-05T13:39:14	Notice	configctl	event @ 1772717954.31 msg: Mar 5 13:39:14 OPNSense.internal config(78808): config-event: new_config/conf/backup/config-1772717954.3024.xml
2026-03-05T13:39:14	Notice	opnsense	/usr/local/sbin/pluginctl: plugins_configure trust_reload (execute task : system_trust_configure(1))
2026-03-05T13:39:14	Notice	opnsense	/usr/local/sbin/pluginctl: plugins_configure trust_reload (1)
2026-03-05T13:39:08	Notice	dhclient	dhclient-script: Creating resolv.conf
2026-03-05T13:39:08	Notice	dhclient	dhclient-script: Reason RENEW on em2 executing
2026-03-05T13:38:06	Error	opnsense	LDAP bind error [80090308: LdapErr: DSID-0C0903A9, comment: AcceptSecurityContext error, data 52e, v1db1; Invalid credentials]
2026-03-05T13:37:51	Error	opnsense	LDAP bind error [80090308: LdapErr: DSID-0C0903A9, comment: AcceptSecurityContext error, data 52e, v1db1; Invalid credentials]
2026-03-05T13:37:04	Notice	configctl	event @ 1772717823.49 exec: system event config_changed response: OK

Création des groupes

Afin que le portail captif fonctionne il faut que les utilisateurs soient dans des groupes, OPNSense est capable de faire le lien entre les groupes auxquels appartient l'utilisateur LDAP et ses propres groupes mais il faut pour cela créer des groupes avec **EXACTEMENT** la même syntaxe que les groupes LDAP.

Dans System → Access → Groups cliquer sur + Add et créer un groupe **Profs**, un groupe **Eleves** et un groupe **Administratifs**

Group name	Privileges	Members	Description	Source Networks	Commands
☐ admins	All pages	root	System Administrators		 
☐ Eleves		jatest	Eleves LDAP		 
☐ Profs		leprof	Profs LDAP		 

Edit Group

Defined By: user

full help

GID: 2002

Group name: Administratifs

Description: Administratifs LDAP
You may enter a description here for your reference (not parsed).

Privileges: Nothing selected
 Clear All Select All

Members: Nothing selected
 Clear All Select All
List of users that are a member of this group

Source Networks:
 Clear All Copy Paste Text
List of networks which constraint the membership of this group to their location (e.g. prevent a non admin user from logging in via internet)

Cancel Save

Tester le bon fonctionnement de la configuration avec System → Access → Tester

Authentication Server	SE4AD
Username	jatest
Password	*****
Test	

Vous devez obtenir ceci :

```
User: jatest authenticated successfully.

This user is a member of these groups:
Elevés

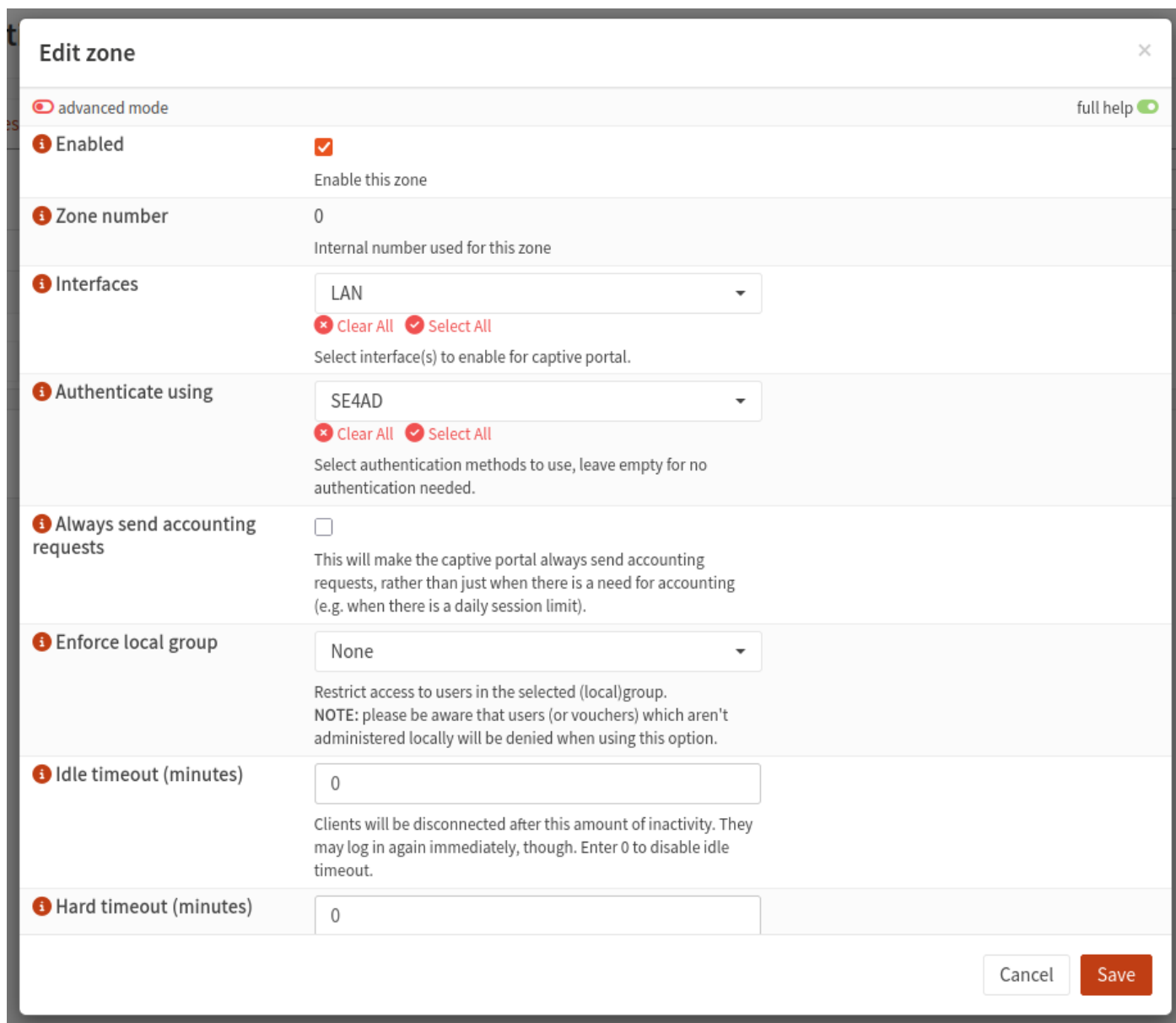
May access the following locations, depending on source address:
Uri                               Networks

Attributes received from server:
dn => CN=jatest,OU=Elevés,OU=utilisateurs,DC=btsciel,DC=vauban,DC=edu
memberof => CN=Elevés,OU=Groups,DC=btsciel,DC=vauban,DC=edu
```

Faites le test avec un user de type :Prof, Eleve et Administratif pour valider.

Configuration du portail captif

Dans *Services* → *Captive Portal* cliquer sur + (Add) et remplir les champs comme dans la capture suivante : Le plus important est : « Authenticate using » où il faut choisir le lien LDAP crée, ici **SE4AD**.



Edit zone [X]

advanced mode [X] full help [X]

Enabled ☒
Enable this zone

Zone number 0
Internal number used for this zone

Interfaces LAN
[X] Clear All [X] Select All
Select interface(s) to enable for captive portal.

Authenticate using SE4AD
[X] Clear All [X] Select All
Select authentication methods to use, leave empty for no authentication needed.

Always send accounting requests ☐
This will make the captive portal always send accounting requests, rather than just when there is a need for accounting (e.g. when there is a daily session limit).

Enforce local group None
Restrict access to users in the selected (local)group.
NOTE: please be aware that users (or vouchers) which aren't administered locally will be denied when using this option.

Idle timeout (minutes) 0
Clients will be disconnected after this amount of inactivity. They may log in again immediately, though. Enter 0 to disable idle timeout.

Hard timeout (minutes) 0

Cancel Save

Portail Captif

Open LDAP sur SE4AD/OPNSense

Edit zone

Hard timeout (minutes)

0

Clients will be disconnected after this amount of time, regardless of activity. They may log in again immediately, though. Enter 0 to disable hard timeout (not recommended unless an idle timeout is set).

Concurrent user logins

If this option is set, users can login on multiple machines at once. If disabled subsequent logins will cause machines previously logged in with the same username to be disconnected.

SSL certificate

Web GUI TLS certificate

If provided, all traffic will be transmitted over an HTTPS connection to protect against eavesdroppers.

Hostname

Hostname (of this machine) to redirect login page to, leave blank to use this interface IP address, otherwise make sure the client can access DNS to resolve this location. When using a SSL certificate, make sure both this name and the cert name are equal.

Allowed addresses

Clear All

Copy

Paste

Text

Avoid authentication for addresses and subnets in this list

Custom template

None

Use custom template package for user login

Description

Test Portal

Description to identify this zone.

Cancel

Save

Validation du fonctionnement

Sur une machine reliée au réseau SE4 lancer un navigateur, vous devez avoir la fenêtre suivante (Après avoir accepté le certificat)



Please sign in

Tester avec un Prof, un Élève et un Administratif.

La configuration du portail est une configuration de base pour valider le concept, il faudra ensuite ajuster les réglages pour l'adapter à nos besoins.